HERIOT WATT UNIVERSITY

School of Mathematical and Computer Sciences

Actuarial Mathematics and Statistics

F78PA

Probability and Statistics A

IN-PERSON EXAMINATION

December Examination and Assessment Diet 2024-2025

Duration: 2 Hours

There are 6 questions in this paper, 60 marks in total.
Candidates must attempt all 6 questions.
To receive full credit you must show your work and explain your answers, i.e. a correct answer without explanation may yield no credit.

Resources Included within Exam Paper: AMS formula sheet
Additional Resources to be Supplied to Candidates: 'New Cambridge Statistical Tables'
Calculator Permitted: Approved electronic calculators may be used

Ethical Commitment: By submitting answers to this paper, I declare that these answers are entirely my own work and have not been shared, in part or in whole or in any form, with any other student, or disseminated in any way.

I also confirm that I have reviewed and understand all University regulations, policies and procedures on assessment misconduct, including but not limited to accessing unauthorised materials during this examination or the use of unauthorised electronic devices.

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Some Continuous Distributions

Name	Notation	p.d.f.	Range	m.g.f.	Mean	Variance
Uniform	$\mathrm{U}(a,b)$	$\frac{1}{b-a}$	$a \le x \le b$	$\frac{e^{bt}-e^{at}}{(b-a)t}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Normal	$\mathrm{N}(\mu,\sigma^2)$	$\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$	$-\infty < x < \infty$	$e^{\mu t+\frac{1}{2}\sigma^2 t^2}$	μ	σ^2
Log Normal	$\mathrm{LogN}(\mu,\sigma^2)$	$\frac{1}{x\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{(\ln x-\mu)^2}{2\sigma^2}\right\}$	$0 < x < \infty$	undefined for $t > 0$	$e^{\mu+\frac{1}{2}\sigma^2}$	$(e^{\sigma^2}-1)(e^{2\mu+\sigma^2})$
Gamma	$\mathrm{Gamma}(\alpha,\lambda)$	$\frac{\lambda^\alpha}{\Gamma(\alpha)}x^{\alpha-1}e^{-\lambda x}$	$0<x<\infty$, $\alpha,\lambda>0$	$\left(\frac{\lambda}{\lambda-t}\right)^\alpha$	$\frac{\alpha}{\lambda}$	$\frac{\alpha}{\lambda^2}$
Exponential	$\mathrm{Exp}(\lambda)$	$\lambda e^{-\lambda x}$	$0<x<\infty$, $\lambda>0$	$\frac{\lambda}{\lambda-t}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Chi-square	χ^2_ν	$\frac{1}{2^{\nu/2}\Gamma(\nu/2)}x^{\nu/2-1}e^{-x/2}$	$0<x<\infty$, $\nu>0$ integer	$(1-2t)^{-\nu/2}$	ν	2ν
Beta	$\mathrm{Beta}(\alpha,\beta)$	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1}$	$0<x<1$, $\alpha,\beta>0$	*omitted	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$
Cauchy	$\mathrm{Cauchy}(\mu,\sigma)$	$\frac{\sigma}{\pi(\sigma^2+(x-\mu)^2)}$	$\sigma > 0$	does not exist		
Pareto	$\mathrm{Pareto}(k,\theta)$	$\frac{k}{\theta}\left(1+\frac{x}{\theta}\right)^{-(k+1)}$	$x>0$, $\theta,k>0$	does not exist	$\frac{\theta}{k-1}$	$\frac{\theta^2 k}{(k-2)(k-1)^2}$
Weibull	$\mathrm{Weibull}(\nu,\lambda)$	$\nu\lambda x^{\nu-1}\exp\left(-\lambda x^\nu\right)$	$x>0$, $\nu,\lambda>0$	*omitted	$\lambda^{-1/\nu}\Gamma\left(1+\frac{1}{\nu}\right)$	*omitted
Multivariate Normal	$\mathrm{N}_p(\boldsymbol{\mu},\boldsymbol{\Sigma})$	$\frac{\exp\left\{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\mathrm{T}}\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right\}}{(2\pi)^{p/2}\det(\boldsymbol{\Sigma})^{1/2}}$	$\mathbf{x}\in\mathbb{R}^p$	$e^{\boldsymbol{\mu}^{\mathrm{T}}\mathbf{t}+\frac{1}{2}\mathbf{t}^{\mathrm{T}}\boldsymbol{\Sigma}\mathbf{t}}$	$\boldsymbol{\mu}$	covariance matrix $\boldsymbol{\Sigma}$

Some Discrete Distributions

Name	Notation	p.m.f.	Range	m.g.f.	Mean	Variance
Discrete Uniform	$\mathrm{DU}(a,b)$	$\frac{1}{b-a+1}$	$x = a,a+1,\ldots,b-1,b$ a,b integers	$\frac{e^{at}-e^{(b+1)t}}{(b-a+1)(1-e^t)}$	$\frac{a+b}{2}$	$\frac{(b-a+1)^2-1}{12}$
Binomial	$\mathrm{Bin}(n,p)$	$\binom{n}{x}p^x(1-p)^{n-x}$	$x = 0,1,2,\ldots,n$ $0<p<1$	$[(1-p)+pe^t]^n$	np	$np(1-p)$
Poisson	$\mathrm{Poi}(\lambda)$	$\frac{e^{-\lambda}\lambda^x}{x!}$	$x = 0,1,2,\ldots$ $\lambda>0$ real	$e^{\lambda(e^t-1)}$	λ	λ
Hyper-geometric	$\mathrm{HypGeo}(n,K,N)$	$\frac{\binom{K}{x}\binom{N-K}{n-x}}{\binom{N}{n}}$	$x = 0,1,2,\ldots,n$ N,K,n integers	*omitted	$\frac{nK}{N}$	$\frac{nK(N-K)(N-n)}{(N)^2(N-1)}$
Geometric	$\mathrm{Geo}(p)$	$p(1-p)^{x-1}$	$x = 1,2,\ldots$ $0<p<1$	$p[e^{-t}-(1-p)]^{-1}$	$1/p$	$(1-p)/p^2$
Negative Binomial	$\mathrm{NegBin}(k,p)$	$\binom{x-1}{k-1}p^k(1-p)^{x-k}$	$x = k,k+1,\ldots$ $0<p<1$	$p^k[e^{-t}-(1-p)]^{-k}$	k/p	$k(1-p)/p^2$

*omitted due to lack of space

Multinomial: multivariate extension of binomial

Probability mass function (p.m.f) is given by

$$f(x_1,x_2,\ldots,x_k)=\frac{n!}{x_1!x_2!\ldots x_k!}p_1^{x_1}p_2^{x_2}\ldots p_k^{x_k},\qquad 0<p_i<1,\forall i=1,2,\ldots,k$$

where $p_1+p_2+\cdots+p_k=1$ and $x_1+x_2+\cdots+x_k=n$.

The m.g.f. is given by

$$[p_1e^{t_1}+p_2e^{t_2}+\cdots+p_ke^{t_k}]^n.$$

Moments: $\mathbb{E}(X_i)=np_i$, $\mathbb{V}\mathrm{ar}(X_i)=np_i(1-p_i)$, $\mathbb{C}ov(X_i,X_j)=-np_ip_j$, $i,j=1,2,\ldots,n$.

1. A medical company is performing a clinical trial to determine the effectiveness of two drugs: drug A and drug B for patients with a particular disease. All the participants of the trial start infected with the disease, and at the end of the trial it is recorded whether the participant recovered, remained the same or deteriorated. They give 20% of the participants a placebo, 40% drug A and the remaining 40% drug B.

- Participants receiving the placebo have a 10% chance of recovering, an 80% chance of remaining the same, and a 10% chance of deteriorating.
- Participants receiving drug A have a 60% chance of recovering, a 30% chance of remaining the same, and a 10% chance of deteriorating.
- Participants receiving drug B have a 70% chance of recovering, a 5% chance of remaining the same, and a 25% chance of deteriorating.

(a) A participant is randomly selected from the trial. If this participant has recovered, what is the probability that they received drug A?

[3 marks]

(b) Of those whose condition **deteriorates** during the trial, 30% need hospital care, while the rest recover after it ends. Is needing hospital care independent of receiving drug B? Justify your answer.

[3 marks]

[Total 6 marks]

2. (a) For a discrete random variable X, the probability mass function is given by:

$$p_X(x) = \begin{cases} \frac{x}{25}, & \text{if } x = 1, 2, 3, 4, 5, \\ \frac{(10-x)}{25}, & \text{if } x = 6, 7, 8, 9, \\ 0, & \text{otherwise.} \end{cases}$$

i. Find the expectation $\mathbb{E}(X)$.

[2 marks]

ii. Determine the probability $\mathbb{P}(X < 9)$.

[2 marks]

(b) In a certain city, the number of customers arriving at a coffee shop during a given hour follows a Poisson distribution with mean λ. The manager has observed that the probability of exactly 2 customers arriving in an hour is equal to the probability of exactly 3 customers arriving. Let Y denote the number of customers arriving at the coffee shop during an hour interval.

i. Determine the variance of Y, $\mathbb{V}\text{ar}(Y)$.

[2 marks]

ii. If the coffee shop can handle up to 3 customers efficiently, what is the probability that the shop will be overwhelmed (i.e., receive more than 3 customers) in a given hour?

[2 marks]

iii. If the mean number of customers λ, were to double, would the probability of receiving more than 3 customers also double? Justify your answer.

[2 marks]

[Total 10 marks]

3. Suppose that X is a random variable with the probability density function

$$f_X(x) = \begin{cases} 3ax^2 + 2bx + 1 & 0 \leq x \leq 1, \\ 0 & \text{otherwise,} \end{cases}$$

where a and b are real numbers with $a > 0$.

(a) Using properties of a density function, show that $b = -a$ for some value $a > 0$.

[2 marks]

(b) By testing the points $0.2, 0.5, 0.7$, or by any other method, explain why $a = 5$ is not a valid choice in this context.

[1 mark]

(c) Now, let $a = 1$ which you may assume to be a valid choice.

(i) Find $F_X(x)$, the cumulative distribution function of X.

[2 marks]

(ii) Calculate $\mathbb{P}(0.4 < X \leq 0.6 \mid X \leq 0.6)$.

[2 marks]

(iii) Let $Y = e^X$. Determine $F_Y(y)$, the cumulative distribution function of Y.

[3 marks]

[Total 10 marks]

4. The cumulative distribution function of a random variable X is defined by:

$$F_X(x) = \begin{cases} 0, & x < 0 \\ \frac{x^2}{2}, & 0 \leq x < 1 \\ \frac{-x^2+4x-2}{2}, & 1 \leq x < 2 \\ 1, & x \geq 2 \end{cases}$$

a) Sketch the graph of $F_X(x)$, and state $\mathrm{Im}X$, the image of X.

[4 marks]

b) Find $\mathbb{P}(0.5 < X \leq 2)$.

[2 marks]

c) Find $f_X(x)$, the probability density function of X.

[3 marks]

[Total 9 marks]

5. Suppose that the joint probability density function of the random variables X and Y be

$$f_{X,Y}(x,y) = \begin{cases} 8xy, & 0 < x < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find the marginal density $f_X(x)$ for X, and the marginal density $f_Y(y)$ for Y.

[4 marks]

(b) Find the expectation of X, $\mathbb{E}[X]$ and of Y, $\mathbb{E}[Y]$.

[2 marks]

(c) Show that the conditional density $f_{Y|X}(y \mid x)$ is given by

$$f_{Y|X}(y|x) = \begin{cases} \dfrac{2y}{(1-x^2)}, & 0 < x < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

[2 marks]

(d) Calculate $\mathbb{E}(Y|X = \frac{1}{2})$.

[3 marks]

(e) Determine the covariance of X, Y, $\mathbb{C}\text{ov}(X, Y)$.

[3 marks]

(f) Intuitively, what does this covariance value imply for the relationship between X and Y?

[1 mark]

[Total 15 Marks]

6. An insurance company handles fraudulent payment claims for a bank, each claim is for £85K. The time T_i between the $(i-1)$-th claim and the i-th claim is exponentially distributed with a rate of 1 claim per hour. The insurance company charges the bank £cK per hour, leading to a profit of:

$$P_n = \sum_{i=1}^{n} (cT_i - 85)$$

(a) Find the value of $\mathbb{E}[P_n]$ and $\mathbb{V}\text{ar}(P_n)$.

[3 marks]

(b) Use the Central Limit Theorem to give the limiting distribution of $\frac{1}{\sqrt{n}}(P_n - \mathbb{E}[P_n])$ as $n \to \infty$.

[2 marks]

(c) Determine the value of c which corresponds to a fair price, i.e. such that $\mathbb{E}[P_n] = 0$ for any n.

[1 mark]

(d) Suppose the company sets $c = 90$. Use the Central Limit Theorem to find the lower bound function, $\ell(n)$, such that

$$\lim_{n\to\infty} \mathbb{P}(\ell(n) \leq P_n) = 0.95.$$

[4 marks]

[Total 10 marks]

END OF PAPER